

Exploratory Data Analysis of Housing Dataset

Hypothesis Testing & Statistical Analysis

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ADTA 5130 – Data Analytics 1

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Introduction

- Real estate pricing is influenced by multiple structural and location-based factors, making accurate prediction a key problem in housing analytics.
- This study performs Exploratory Data Analysis (EDA) on a dataset of 19,432 housing transactions with 16 variables describing property features.
- The objective is to understand how different housing attributes impact sale price variations using statistical methods.
- The analysis combines ANOVA tests and Simple Linear Regression models to evaluate relationships between categorical and numerical variables.
- Key focus areas include property condition, number of floors, living area size, above-ground space, and number of bathrooms in relation to house prices.

Dataset Description

Numerical Variables (9)	
price	Continuous – property sale value (target)
bedrooms	Discrete – number of bedrooms
bathrooms	Continuous – number of bathrooms
sqft_living	Total interior living space (sqft)
sqft_lot	Total lot size (sqft)
sqft_above	Living area above ground (sqft)
sqft_basement	Basement area (sqft)
yr_built	Year the property was built
yr_renovated	Year of last renovation

19,432
Observations

16
Variables

Categorical Variables (7)	
waterfront	Binary – has waterfront view or not
view	Rated 0–4 (view quality)
condition	Rated 1–5 (property condition)
grade	Rated 3–13 (construction quality)
floors	Number of floors (incl. halves: 1, 1.5, 2 ...)
zipcode	Nominal location identifier
date.sold	Date of sale transaction

2
ANOVAs

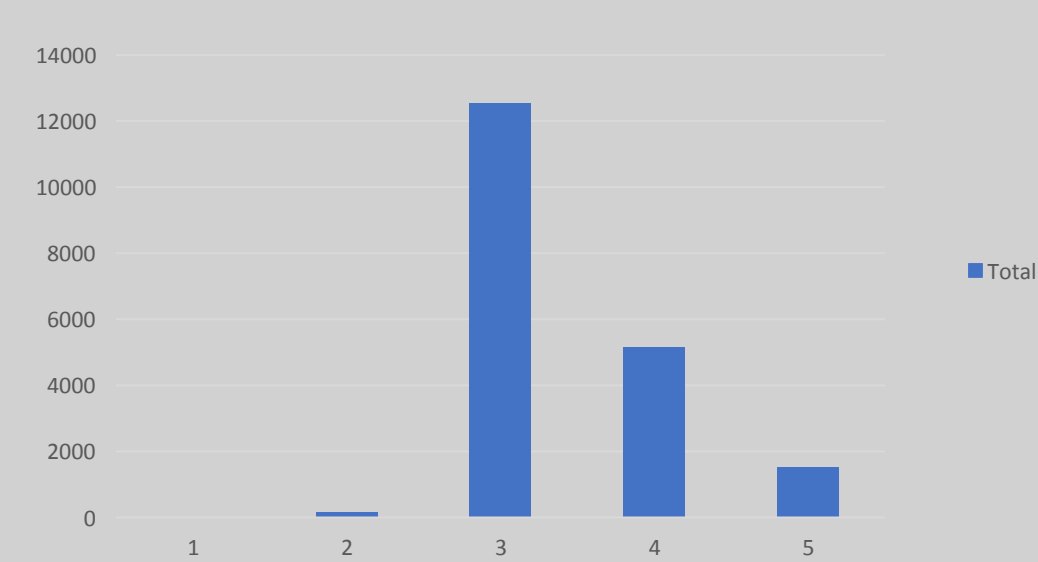
3
Regressions

Exploratory Data Analysis – Distributions

Property Condition & Number of Floors

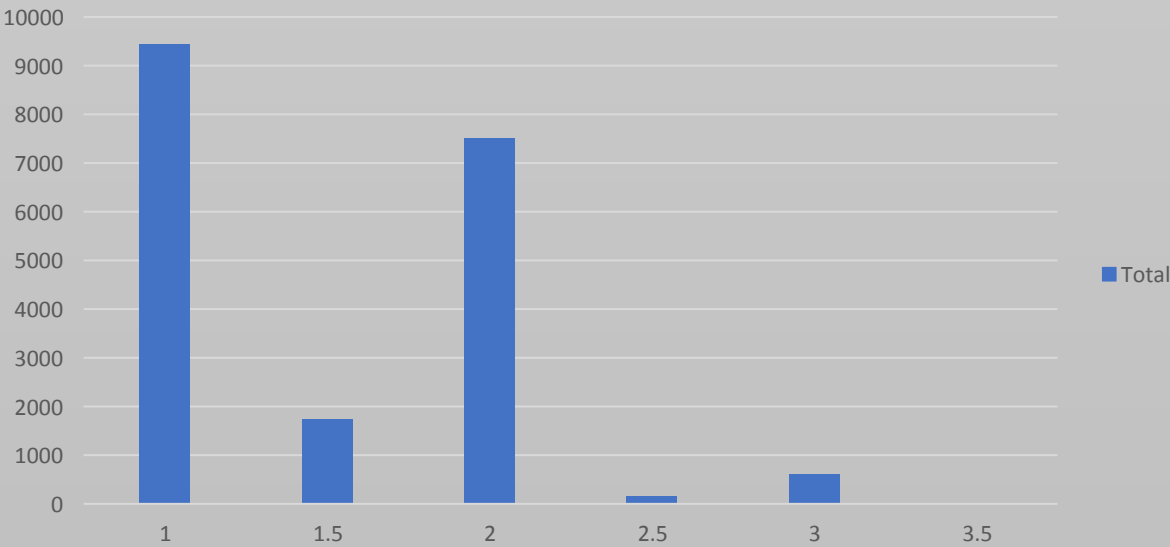
Row Labels	Count of condition
1	27
2	153
3	12556
4	5160
5	1536
Grand Total	19432

Condition Distribution



Row Labels	Count of floors
1	9430
1.5	1727
2	7509
2.5	152
3	606
3.5	8
Grand Total	19432

Floor Distribution



Descriptive Statistics

Key numerical variables – 19,432 observations

	price	bedrooms	bathrooms	sqft_living	sqft_lot	sqft_above	sqft_basement	yr_built	yr_renovated
Mean	556685.992	3.37747	2.1287438	2091.54168	13756.95245	1794.796367	296.745317	1971.014769	82.27850967
Standard Error	2681.40334	0.006736	0.0055376	6.63463481	257.5921279	5.9750269	3.189038445	0.212892288	2.846783047
Median	469000	3	2.25	1920	7500	1560	0	1975	0
Mode	450000	3	2.5	1300	5000	1010	0	2014	0
Standard Deviation	373784.163	0.938932	0.7719304	924.859525	35908.01001	832.9110346	444.5478414	29.67691011	396.8378808
Sample Variance	1.3971E+11	0.881594	0.5958766	855365.141	1289385183	693740.7916	197622.7833	880.7189934	157480.3037
Kurtosis	34.4822041	52.42479	1.2847665	5.24822677	288.8551327	3.274862133	2.71457009	-0.647562162	19.31388775
Skewness	4.01908526	2.111699	0.5103564	1.46838521	13.57681242	1.426900015	1.560814784	-0.495415838	4.616311249
Range	7620000	33	8	13170	1164274	9040	4820	115	2015
Minimum	80000	0	0	370	520	370	0	1900	0
Maximum	7700000	33	8	13540	1164794	9410	4820	2015	2015
Sum	1.0818E+10	65631	41365.75	40642838	267325100	34876483	5766355	38300759	1598836
Count	19432	19432	19432	19432	19432	19432	19432	19432	19432

Correlation with Price

0.704

sqft_living

0.607

sqft_above

0.526

bathrooms

0.313

bedrooms

0.093

sqft_lot

Price is right-skewed (skewness 4.02) — mean (\$556K) is pulled above median (\$469K) by a few luxury homes.

Research Questions & Hypotheses

5 hypotheses tested via ANOVA and Simple Linear Regression · $\alpha = 0.05$

RQ1

Does property condition rating influence house sale price?

Test: One-Way ANOVA + Tukey HSD

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

H_a : At least one group mean differs

RQ2

Does the number of floors influence house sale price?

Test: One-Way ANOVA + Tukey HSD

$$H_0: \mu_1 = \mu_{1.5} = \mu_2 = \mu_{2.5} = \mu_3 = \mu_{3.5}$$

H_a : At least one group mean differs

RQ3

Does sqft_living have a significant linear relationship with price?

Test: Simple Linear Regression

$$H_0: \beta = 0$$

$$H_a: \beta \neq 0$$

RQ4

Does sqft_above have a significant linear relationship with price?

Test: Simple Linear Regression

$$H_0: \beta = 0$$

$$H_a: \beta \neq 0$$

RQ5

Does the number of bathrooms have a linear effect on price?

Test: Simple Linear Regression

$$H_0: \beta = 0$$

$$H_a: \beta \neq 0$$

RQ1: Effect of Property Condition on House Price

One-Way ANOVA + Tukey's HSD Post-Hoc Test

$F(4, 19427) = 37.02$ · $p\text{-value} = 6.91 \times 10^{-31}$ · Decision: Reject H_0 — significant differences exist across condition groups

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
c1	27	9731950	360442.5926	74893227443		
c2	153	52445564	342781.4641	64776097134		
c3	12556	7028796205	559795.8112	1.37941E+11		
c4	5160	2750687717	533079.0149	1.32949E+11		
c5	1536	975860751	635326.0098	1.72465E+11		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	2.05364E+13	4	5.1341E+12	37.01952986	6.90567E-31	2.372389414
Within Groups	2.69426E+15	19427	1.38686E+11			
Total	2.71479E+15	19431				

comparison	diff	std error	margin of error	lwr	upr	p adj
c2-c1	-17661.13	54968	212066.56	-229727.69	194405.43	0.5805
c3-c1	199353.22	50732.45	195725.81	3627.41	395079.03	0.0472202
c4-c1	172636.42	50810.41	196026.56	-23390.14	368662.99	0.1191
c5-c1	274883.42	51121.47	197226.62	77656.8	472110.04	0.0148318
c3-c2	217014.35	21418.33	82631.92	134382.42	299646.27	< 0.0001
c4-c2	190297.55	21602.33	83341.8	106955.75	273639.35	0.0009523
c5-c2	292544.55	22324.14	86126.55	206418	378671.09	< 0.0001
c4-c3	-26716.8	4354.45	16799.48	-43516.28	-9917.32	0.008085
c5-c3	75530.2	7118.14	27461.77	48068.42	102991.97	< 0.0001
c5-c4	102246.99	7654	29529.14	72717.85	131776.14	< 0.0001

- There is a statistically significant difference in mean house prices across different property condition levels ($F = 37.02$, $p < 0.05$).
- We reject the null hypothesis, meaning condition level does affect house price.
- Houses in higher condition (especially condition 5) generally have higher average prices.
- Tukey's HSD shows that the biggest price gaps are between extreme conditions (e.g., 1 vs 5).
- Overall, property condition has a meaningful but non-linear impact on house price.

Conclusion: Better property condition → higher price. Effect strongest between extreme conditions (1 vs 5). c2 vs c1 not significant.

RQ2: Effect of Number of Floors on House Price

One-Way ANOVA + Tukey's HSD Post-Hoc Test

$F(5, 19426) = 366.30$ · $p\text{-value} < 0.001$ · Decision: Reject H_0 — significant differences exist across floor groups

Anova: Single Factor						
SUMMARY						
Groups	Count	Sum	Average	Variance		
1	9430	4286832774	454595.204	7.2627E+10		
1.5	1727	1000500750	579328.7493	9.1781E+10		
2	7509	5006060616	666674.7391	1.9282E+11		
2.5	152	164906785	1084913.059	7.6478E+11		
3	606	351754762	580453.4026	1.1212E+11		
3.5	8	7466500	933312.5	9.5769E+11		
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	2.33899E+14	5	4.67798E+13	366.296784	0	2.21456
Within Groups	2.4809E+15	19426	1.2771E+11			
Total	2.71479E+15	19431				
comparison	diff	std error	margin of error	lwr	upr	p adj
1.5-1	124733.55	6614.08	25517.1	99216.44	150250.65	< 0.0001
2-1	212079.54	3908.36	15078.45	197001.09	227157.98	< 0.0001
2.5-1	630317.85	20660.84	79709.51	550608.35	710027.36	< 0.0001
3-1	125858.2	10589.75	40855.25	85002.95	166713.45	< 0.0001
3.5-1	478717.3	89379.23	344825.06	133892.23	823542.36	0.0150835
2-1.5	87345.99	6743.76	26017.43	61328.56	113363.42	< 0.0001
2.5-1.5	505584.31	21379.27	82481.23	423103.08	588065.54	< 0.0001
3-1.5	1124.65	11930.88	46029.32	-44904.67	47153.97	0.6146
3.5-1.5	353983.75	89548.03	345476.3	8507.45	699460.05	0.0463406
2.5-2	418238.32	20702.72	79871.08	338367.24	498109.4	< 0.0001
3-2	-86221.34	10671.23	41169.59	-127390.93	-45051.74	0.0017067
3.5-2	266637.76	89388.92	344862.45	-78224.69	611500.21	0.1813
3-2.5	-504459.66	22923.13	88437.44	-592897.1	-416022.21	< 0.0001
3.5-2.5	-151600.56	91662.28	353633.08	-505233.64	202032.52	0.3806
3.5-3	352859.1	89929.12	346946.54	5912.56	699805.64	0.0474381

Conclusion: More floors → higher price. 2.5-floor homes show the largest premium (\$1.08M avg). Relationship is not perfectly linear.

- There is a statistically significant difference in house prices across floor categories ($F = 366.30$, $p < 0.05$).
- We reject the null hypothesis, meaning number of floors influences house price.
- Houses with 2 or more floors tend to have higher prices than single-floor houses.
- Tukey's HSD shows clear price differences between 1-floor and multi-floor houses.
- However, the relationship is not perfectly proportional due to uneven data distribution across floor types.

RQ3: Living Area (sqft_living) vs House Price

Simple Linear Regression · Reject $H_0: \beta \neq 0$

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.704347465
R Square	0.496105351
Adjusted R Square	0.496079417
Standard Error	265339.524
Observations	19432

ANOVA					
	df	SS	MS	F	Significance F
Regression	1	1.34682E+15	1.34682E+15	19129.64742	0
Residual	19430	1.36797E+15	70405063011		
Total	19431	2.71479E+15			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-38699.98787	4706.780539	-8.222178101	2.12281E-16	-47925.68291	-29474.29283	-47925.68291	-29474.29283
sqft_living	284.6636928	2.058157454	138.3099686	0	280.629527	288.6978586	280.629527	288.6978586

Multiple R

0.704

Strong positive correlation

F-statistic

19,129.65

p < 0.001 — model highly significant

R²

0.496

49.6% of price variation explained

Coefficient

\$284.66

per additional sqft of living area

Regression Equation & Interpretation

Price = −\$38,700 + \$284.66 × sqft_living

- For every additional 1 sqft of living area, house price increases by approximately \$284.66 on average.
- The intercept (−\$38,700) has no practical meaning as a house price cannot be negative — it is a mathematical artifact.
- R² = 0.496: sqft_living alone explains 49.6% of the variation in house price, making it the strongest single predictor.
- The model is statistically significant (F = 19,129.65, p < 0.001) — H₀ is rejected.

sqft_living is the strongest predictor of price in this dataset — R² = 49.6%

RQ4 : Sqft_Above and House price

Simple Linear Regression · H_0 rejected ($\beta \neq 0$)

RQ4 · sqft_above vs Price

Multiple R

0.607

R^2

36.9%

F-stat

11,356.67

Coeff

+\$272.56/sqft

Price = \$67,491 + \$272.56 × sqft_above

- Each additional sqft above ground → +\$273 in price.
- $R^2 = 0.369$: explains 36.9% of price variance.
- Slightly weaker predictor than sqft_living (49.6%).
- Intercept (\$67,491) is baseline — limited practical meaning.

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.607357129								
R Square	0.368882682								
Adjusted R Square	0.3688502								
Standard Error	296952.5926								
Observations	19432								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	1.00144E+15	1.00144E+15	11356.66904	0				
Residual	19430	1.71335E+15	88180842247						
Total	19431	2.71479E+15							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	67491.45704	5060.655363	13.33650529	2.13719E-40	57572.13688	77410.7772	57572.13688	77410.7772	
sqft_above	272.5626949	2.557648918	106.5676735	0	267.5494828	277.5759069	267.5494828	277.5759069	

Both RQ4 and RQ5: H_0 rejected · sqft_above ($R^2=36.9\%$) is a stronger predictor than bathrooms ($R^2=27.6\%$)

RQ5: Bathrooms and House Price

Simple Linear Regression · Both H_0 rejected ($\beta \neq 0$)

SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.525698958								
R Square	0.276359394								
Adjusted R Square	0.276322151								
Standard Error	317975.4106								
Observations	19432								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	1	7.50259E+14	7.50259E+14	7420.3451	0				
Residual	19430	1.96454E+15	1.01108E+11						
Total	19431	2.71479E+15							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	14805.82052	6691.388473	2.212667906	0.026932131	1690.123086	27921.51796	1690.123086	27921.51796	
bathrooms	254553.9603	2955.070227	86.14142499	0	248761.7683	260346.1523	248761.7683	260346.1523	

RQ5 · Bathrooms vs Price

Multiple R

0.526

R²

27.6%

F-stat

7,420.35

Coeff

+\$254,554/bath

Price = \$14,806 + \$254,554 × bathrooms

- Each additional bathroom → +\$254,554 in price on average.
- R² = 0.276: explains 27.6% of price variance.
- Significant but weakest of the three regression models.
- Intercept (\$14,806) has limited practical interpretation.

Both RQ4 and RQ5: H_0 rejected · sqft_above ($R^2=36.9\%$) is a stronger predictor than bathrooms ($R^2=27.6\%$)

Conclusion

All 5 hypotheses tested · $\alpha = 0.05$

RQ1	ANOVA	Reject H ₀	Property condition significantly affects price (F=37.02, p<0.001). Higher condition → higher price. Effect strongest between extremes (Condition 1 vs 5). Condition 5 averages \$635K vs \$343K for Condition 2.
RQ2	ANOVA	Reject H ₀	Number of floors significantly affects price (F=366.30, p<0.001). 2.5-floor homes average \$1.08M — largest premium. Relationship is not perfectly linear (fewer observations at extremes).
RQ3	Regression	Reject H ₀	sqft_living is the strongest predictor (R ² =49.6%). Each additional sqft adds ~\$285 to price. Model: Price = −\$38,700 + \$284.66 × sqft_living. Highly significant (F=19,129.65).
RQ4	Regression	Reject H ₀	sqft_above explains 36.9% of price variation. Each additional sqft adds ~\$273. Slightly weaker than total living area. Model: Price = \$67,491 + \$272.56 × sqft_above.
RQ5	Regression	Reject H ₀	Each bathroom adds ~\$254,554 to price on average (R ² =27.6%). Significant but weakest predictor among the three regression models. Model: Price = \$14,806 + \$254,554 × bathrooms.

Overall finding: House prices are primarily driven by interior space (sqft_living, R²=49.6%) and structural attributes. All 5 null hypotheses rejected at $\alpha = 0.05$.